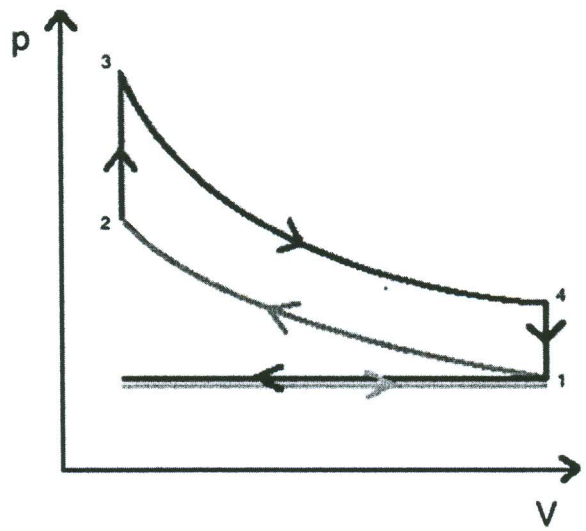
Physical Chemistry
Thermodynamics 1//23/15

1. (50 pts) The cycle involved in the operation of an internal combustion engine is called the Otto cycle. Air can be considered to be the working substance and can be assumed to be an ideal gas. Part of the cycle consists of the following steps: (I) reversible adiabatic compression from 1 to 2, (II) reversible constant-volume pressure increase from 2 to 3 due to combustion of a small amount of fuel, (III) reversible adiabatic expansion from 3 to 4, and (IV) reversible and constant-volume pressure decrease back to state 1. (For a reversible adiabatic process, $TV^{\gamma-1}$ = constant, where $\gamma = C_{p,m}/C_{v,m}$)



(a) Show that the maximum theoretical efficiency of this engine $\eta = 1-T_1/T_2 = 1-T_4/T_3$.
(b) Assume that the engine is operating in the Otto cycle and that $C_{v,m} = 5R/2$ for the fuel-air mixture initially at 298 K (T_1). The mixture is compressed by a factor of 7.5 in the adiabatic compression step. What is the maximum theoretical efficiency of this engine?
(c) How much would be the efficiency if the compress ratio could be increased to 25. Do you see a problem in doing so?

2. (50 pts) One mole of $H_2O(l)$ is supercooled to -2.25 °C at 1 bar pressure. The freezing temperature of water at this pressure is 0.00 °C. The transformation $H_2O(l) \rightarrow H_2O(s)$ is suddenly observed to occur. By calculating ΔS, $\Delta S_{surroundings}$, and ΔS_{total}, verify that this transformation is spontaneous at -2.25 °C. The heat capacities are given by $C_p(H_2O(l)) = 75.3$ J $K^{-1}mol^{-1}$ and $C_p(H_2O(s)) = 37.7$ J $K^{-1}mol^{-1}$, and $\Delta H_{fusion} = 6.008$ kJ mol^{-1} at 0.00 °C. Assume that the surrounding are at -2.25 °C.